Predictive Traffic Telemetry and Crash Forensics

Using Python

1. Introduction

This project focuses on analyzing Motor Vehicle Crash Data from 2024 to 2026 using Python to understand road accident patterns, identify major contributing factors, and evaluate accident severity. The dataset contains information about crash dates and times, geographic locations, vehicle types, contributing factors, and the number of people injured or killed.

The analysis focuses on:

- Accident trends across different dates and years
- Crash patterns based on time periods (Morning, Afternoon, Evening, and Night)
- Geographic distribution of accidents by borough and location
- Major contributing factors associated with vehicle crashes
- Vehicle types involved in accidents
- Number of pedestrian, cyclist, and motorist injuries
- Crash severity and fatality patterns
- Identification of high-risk locations and accident trends

Objective

The main objective of this project is to analyze motor vehicle crash data to identify accident patterns, major contributing factors, high-risk locations, and the severity of road accidents.

Using Python, Pandas, NumPy, Matplotlib, and Seaborn, the project aims to clean and transform raw crash records into meaningful insights that can support better road safety awareness, accident prevention, and data-driven decision-making.

Key Objectives – Motor Vehicle Crash Data Analysis (2024–2026)

- Identify accident trends across different years and reporting periods.
- Analyse accident frequency based on Morning, Afternoon, Evening, and Night.
- Identify the major contributing factors responsible for motor vehicle crashes.

Analyse the number of persons injured and killed in road accidents. • Compare crash severity across different vehicle types.

• The relationship between accident frequency, time periods, and crash severity.

1.Dataset information

Attribute	Details
Data Source	NYC Open Data
Domain	Transportation / Road Safety
Geographic Coverage	New York City, USA
Year	2024–2026
File Format	CSV / Excel
Environment	Google Colab
Programming Language	Python
Libraries	Pandas, NumPy, Matplotlib, Seaborn

2. Dataset structure

Column group	Examples of fields	Purpose
Date and time	CRASH DATE, CRASH TIME	Identify when the crash occurred.
Geographic details	BOROUGH, ZIP CODE, LATITUDE, LONGITUDE, LOCATION	Describe where it occurred.
Road details	ON STREET NAME, CROSS STREET NAME, OFF STREET NAME	Record road or intersection information
Casualties	NUMBER OF PERSONS INJURED/KILLED; pedestrian, cyclist and motorist counts	Measure reported injuries and deaths.

Crash factors	CONTRIBUTING FACTOR VEHICLE 1 and 2	Capture reported contributing factors.
Vehicle details	VEHICLE TYPE CODE 2, 3, 4 and related vehicle fields	Describe involved vehicle types.

3.Data types to check

Field type	Suggested type / treatment
CRASH DATE	Convert to datetime after confirming the date format.
CRASH TIME	Parse as time or retain as text initially; validate values such as HH: MM.
ZIP CODE	Keep as string to preserve leading zeros and avoid treating it as a measurement.
LATITUDE / LONGITUDE	Numeric (float); check valid geographic ranges and missing values.
Injury / fatality counts	Numeric integers; inspect blanks, negative values, and implausible counts.
Borough, street, contributing factor, vehicle type	Categorical/text; standardize whitespace and labels such as “Unspecified”.

USER INTERFACE DIAGRAM

Data
collection

Data load

Data set Information

Data Cleanning

Feature
Engineering(EDA)

Data visualiaztion

Crash count

Crash Year count

Monthly wise crash
Time period

Number of vehicle crash

Number of
and vehicle type
Motorist killed

DATA LOAD

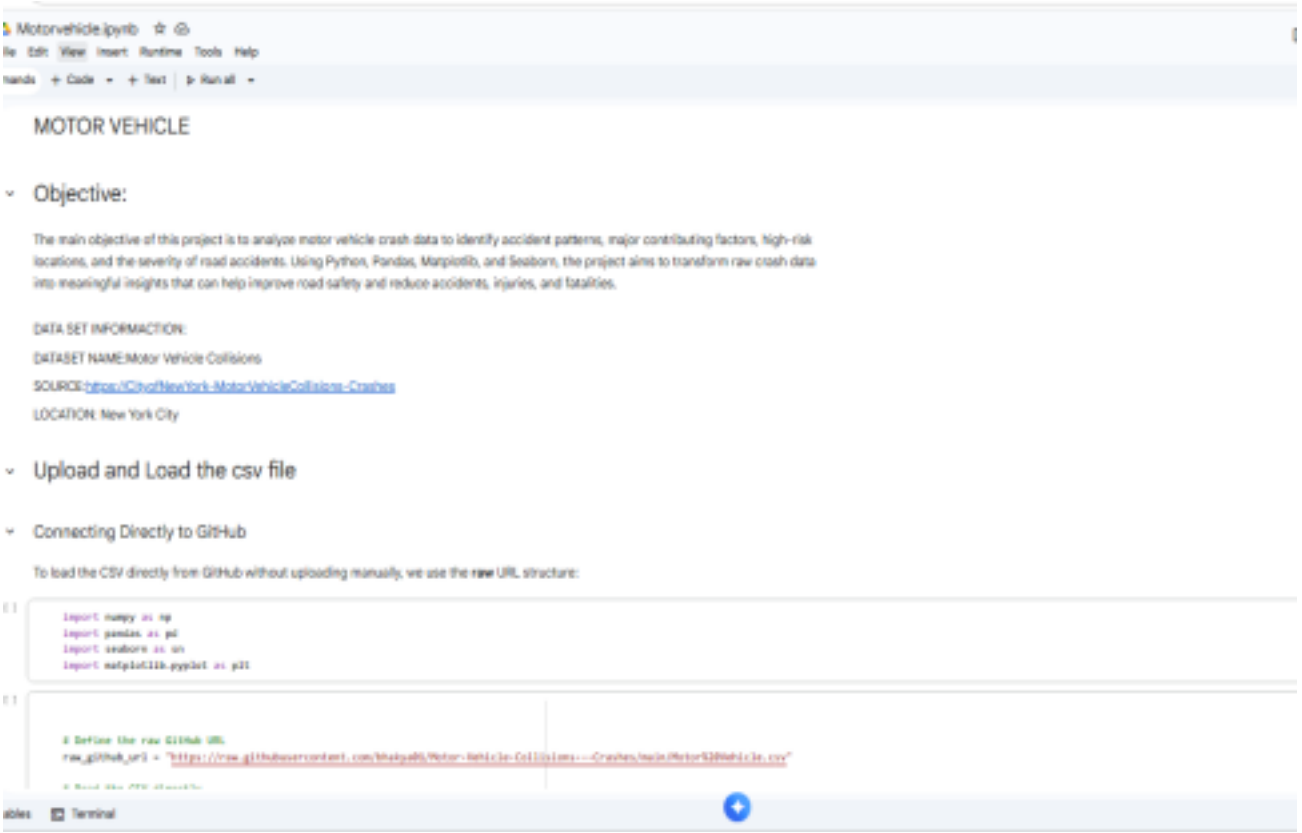
Connecting Directly to GitHub

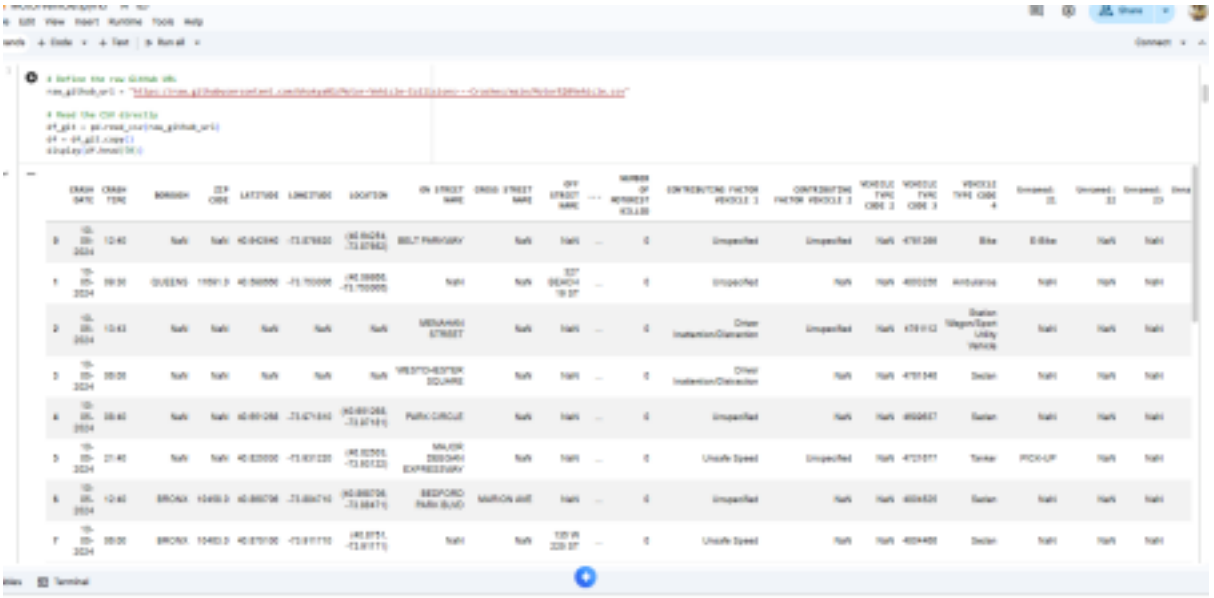
To load the CSV directly from GitHub without uploading manually, we use the raw URL structure:

DATASET NAME: NYC ROAD SAFETY INTELLIGENCE

SOURCE : https://CityofNewYork-MotorVehicleCollisions-Crashes

LOCATION: New York City

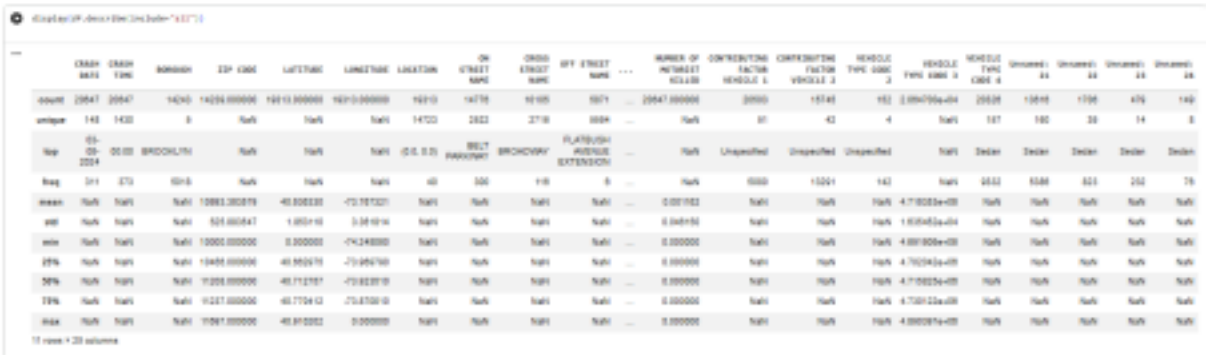




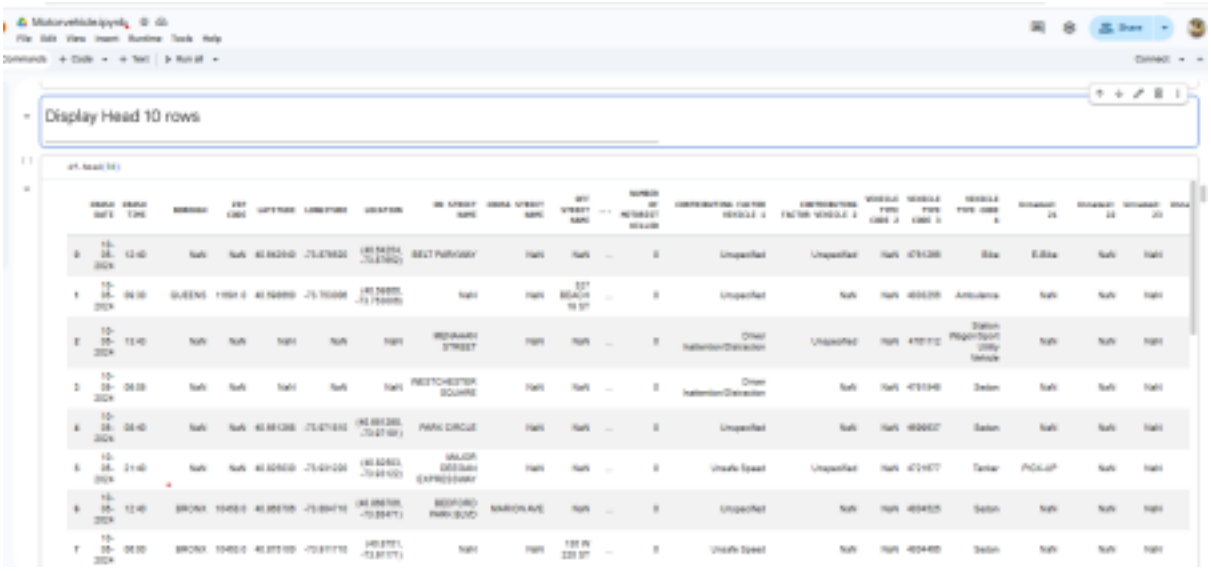
DATA SET INFORMATION



Generate descriptive statistics

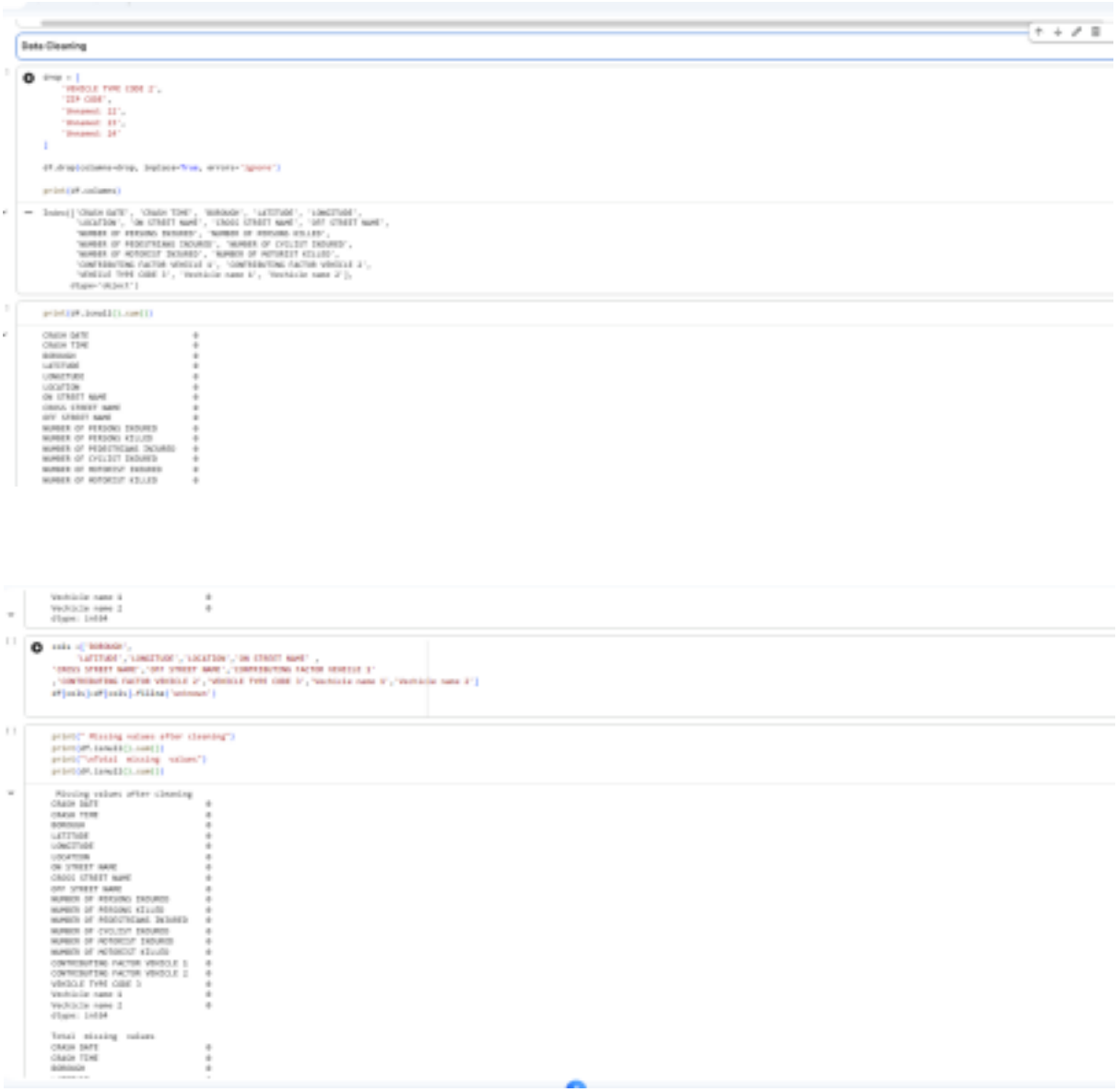


DISPLAY DATA SET FIRST 10 ROW



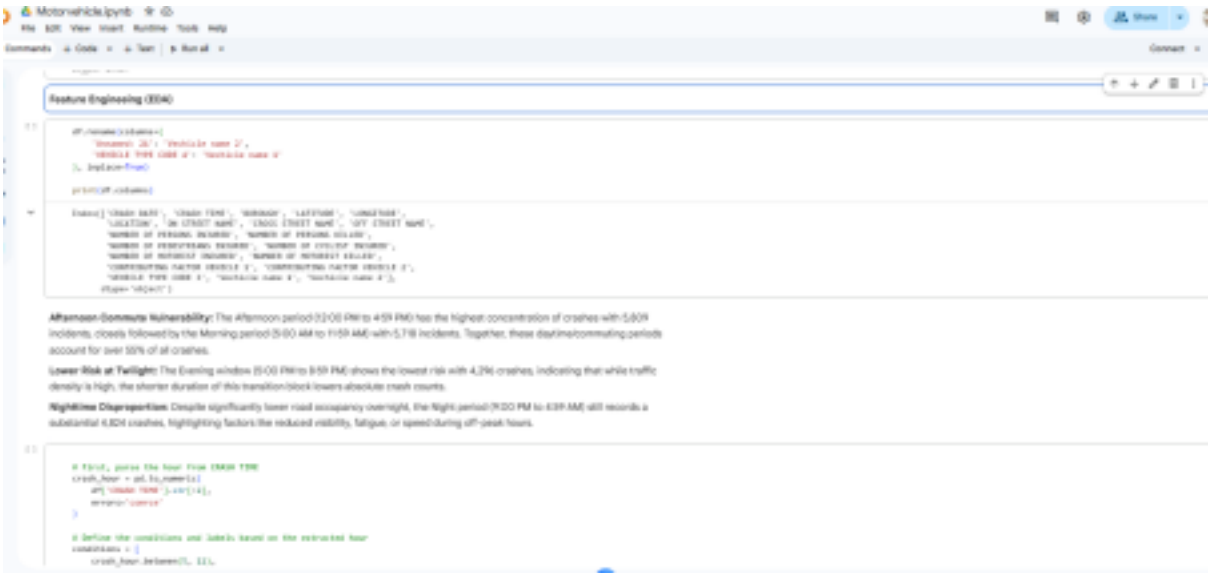
DATA CLEANNING

- Data set cleaning **:** In my NYC ROAD SAFETY INTELLIGENCE I drop my (vehicle type code 2,zip code , unnamed 22,unnamed 23,unnamed 24 has been ignore
- I cleaned using is null
- Missing value after cleaning and Total missing values



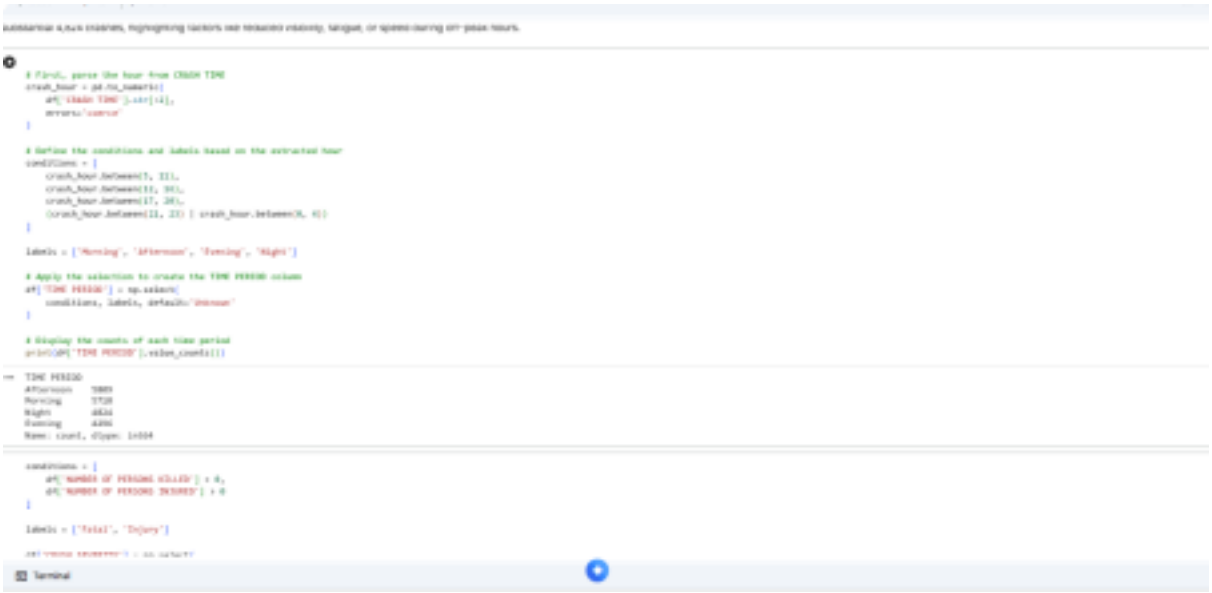
FEATURE ENGINEERING(EDA)

RENAME : In NYC ROAD SAFETY INTELLIGENCE Data set record I rename columns (Unamed23 has Vechicle name 2 and VECHICLETYPE CODE 4 has Vechicle name1)



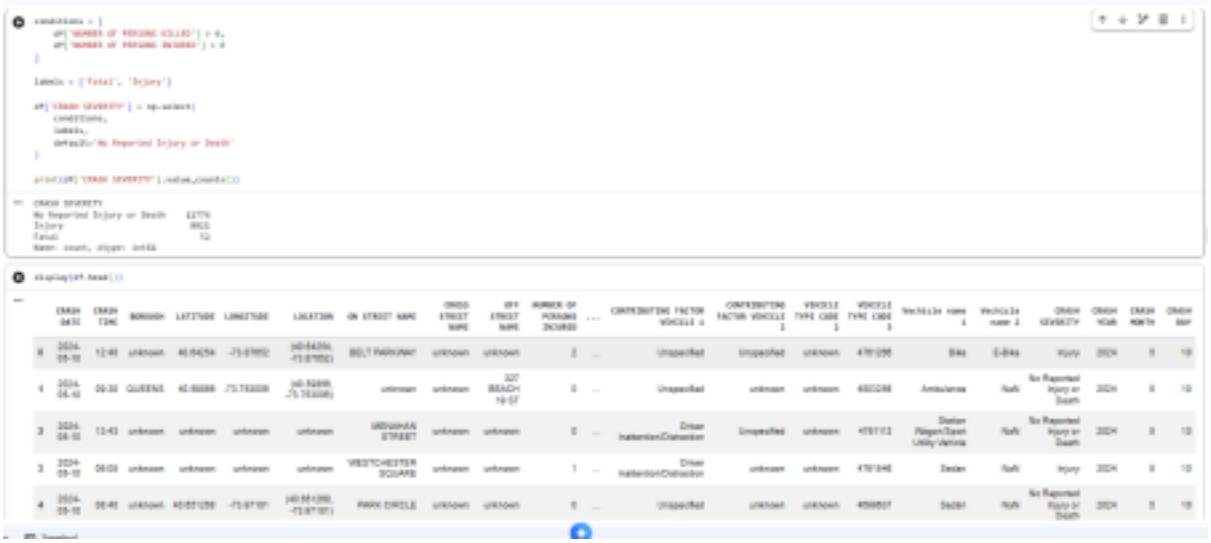
Crash hour :crash time has been splitted in 4 parts.

S.No	Crash Hour (24-Hour Format)	Time Period
1.	05:00 – 11:59	Morning
2.	12:00 – 16:59	Afternoon
3.	17:00 – 20:59	Evening
4.	21:00 – 23:59 and 00:00 – 04:59	Night



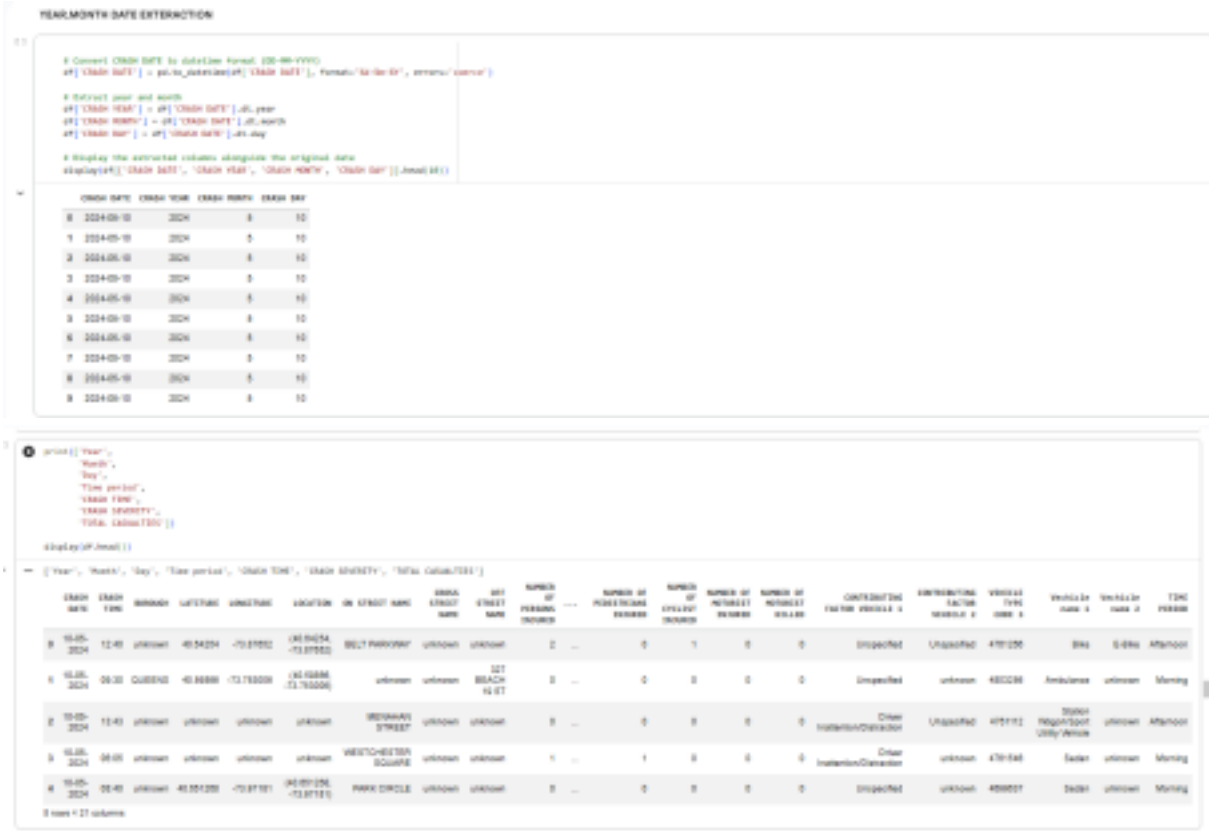
Crash severity: Number of persons killed and Number of persons injured has been separated into 3 types

- NoReported Injury or death:11774
- Injury:8825
- Fatal:52



Crash Date has been extraction

- Crash year (2024-2026)
- Crash month (1-12)
- Crash date (1-30)



Data visualizations:

Number of crash year and crash count in year of 2024 the crash count 20000 has been increased and In the year of 2025 in 350 has been less them when compared to last one year an

Key Insights – Year-wise Crash Analysis

1. Highest Crash Count: The year 2024 recorded approximately 20,000 crashes, indicating a high number of reported road accidents.

2. Decrease in 2025: Crash incidents decreased by approximately 350 compared with 2024, showing a slight reduction in reported crashes.

3. No Recorded Crashes in 2026: The dataset shows no crash records for 2026. This may indicate incomplete data rather than zero accidents.

4. Overall Trend: The comparison highlights a small decline in recorded crashes from 2024 to

2025. there is no crash count in 2026



Monthly wise crash distribution key insight

- In this monthy wise crash distibution in(May to June)in spring and summer month
- (October to November) is holiday is increased trips
- Lower Accident count(Feb and Jan) winter weather

Key Insights – Crash Time Period vs Crash Count

- Afternoon Peak: Afternoon recorded the highest number of crashes, with 5,809 incidents, indicating a higher crash frequency during this period.
- Morning Traffic: Morning recorded 5,718 crashes, making it the second-highest accident period.
- Evening Decline: Evening recorded 4,290 crashes, showing the lowest number of incidents among the four time periods. • Night Accidents: Night recorded 4,824 crashes, indicating that accidents continue to occur during late hours. • Overall Pattern: Morning and afternoon together account for a larger share of reported crashes than evening and night.



Bar chart crash time period and crash count

- Accidents has been peak in Afteroon(5,809)and Morning(5,718) Because its heavy traffic
- Lower Accidents in Evenning(4,290)and Night(4,824) lower collsion

Key Insights – Crash Time Period vs Crash Count

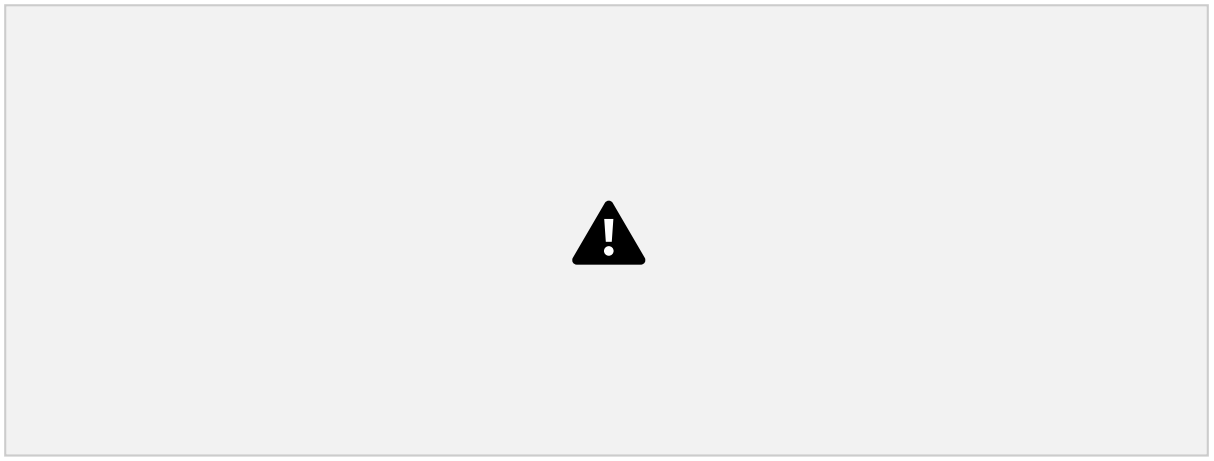
- Afternoon Peak: Afternoon recorded the highest number of crashes, with 5,809 incidents.
- Morning Traffic: Morning recorded the second-highest number, with 5,718 crashes.

- Lowest Crash Count: Evening recorded the lowest number of crashes, with 4,290 incidents.
- Night-time Crashes: Night recorded 4,824 crashes, showing that road accidents also occur during late hours. • Traffic Impact: Higher crash counts during morning and afternoon may be associated with heavy traffic and peak-hour commuting



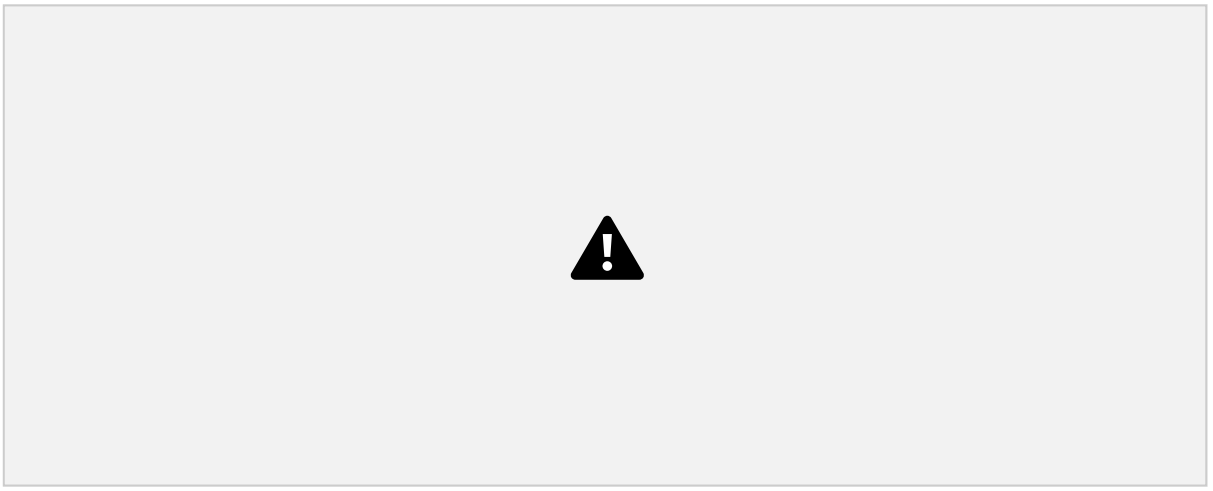
Number of vehicle crash and vehicle type

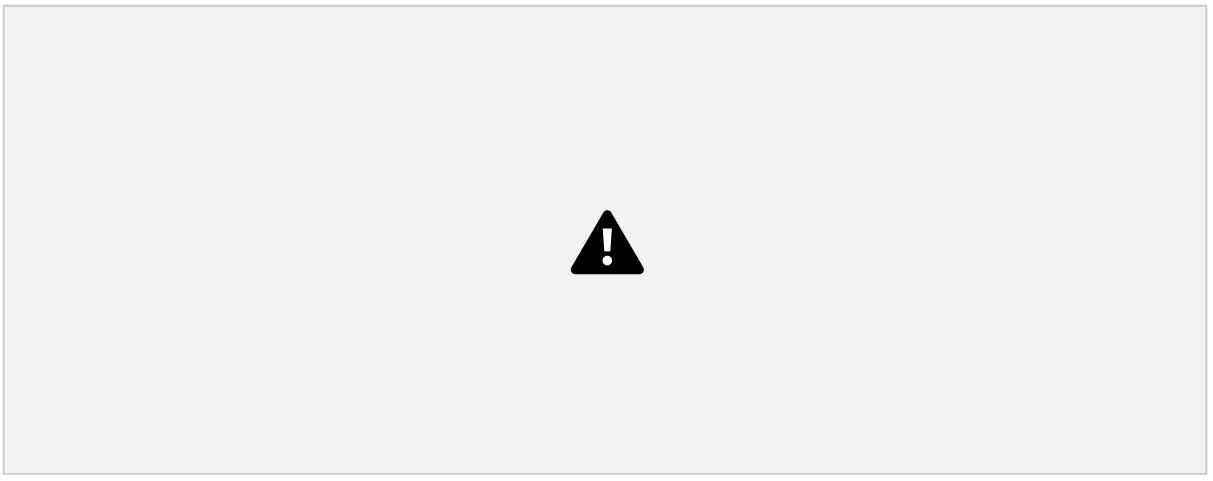
- Sedan is above 8000,station wagons utility vehicle less than 7000
- Tax,pick up truck and bus(1500)
- Bike, box truck, motorcycle(500)
- Moped, Tractor ,Truck Diesel(100)



Pivot table

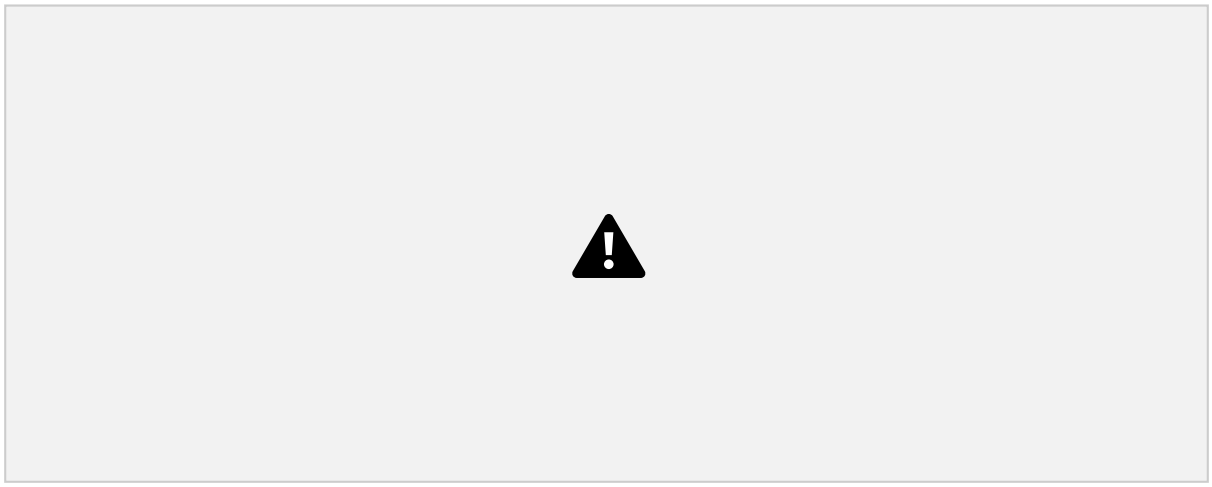
• Most Dangerous Collision Dynamics in Sport Utility Vehicles (1,211 injuries) and sedans(1794 injuries) • . Because of mutual collisions dominate the overall safety metrics, producing the largest clusters of motorist injuries in the entire dataset.





GROUP BY

Vechicle name 1,Vechicle name 2 has been grouped by Number of Motorist killed

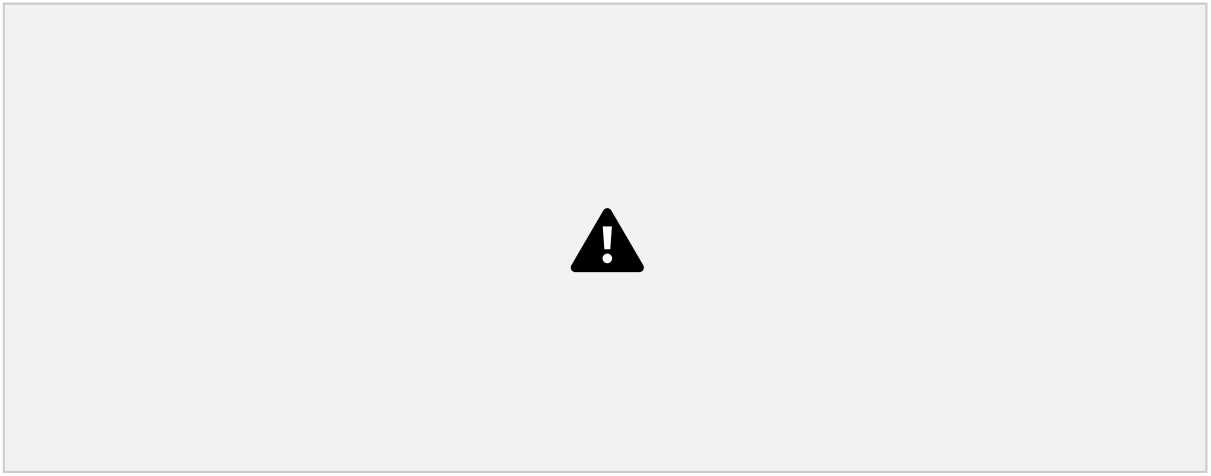


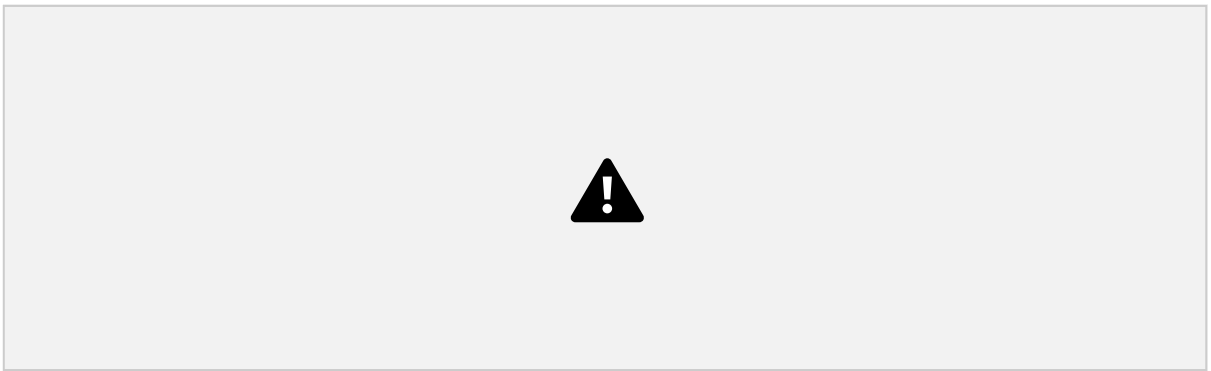
Bar chat

• Highest Fatalities: Sedan , Station Wagon/Sport Utility Vehicle recorded the highest number, with 5 motorist deaths • Other Vehicle Tractor Truck Diesel, Motorcycle recorded 1 death.
Motorcycle-related Fatalities: Motor scooter , Bus and Motorcycle , Tractor Truck Diesel each recorded 1 death • Low Fatality Counts: Most of the displayed vehicle combinations recorded only 1 death, while some recorded zero.

Draw back in Sedan

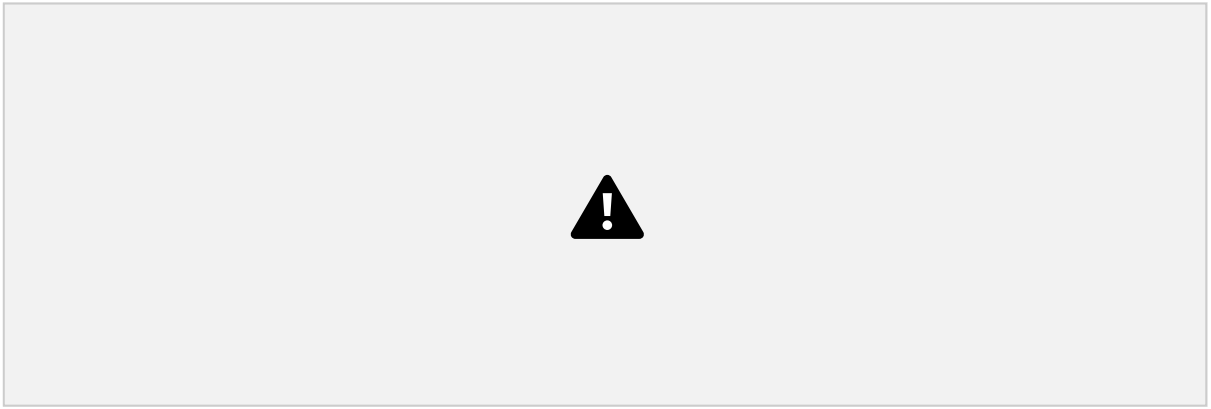
Possible Risk Factors: Differences in vehicle size, weight, visibility, collision speed and impact direction may influence crasseverity

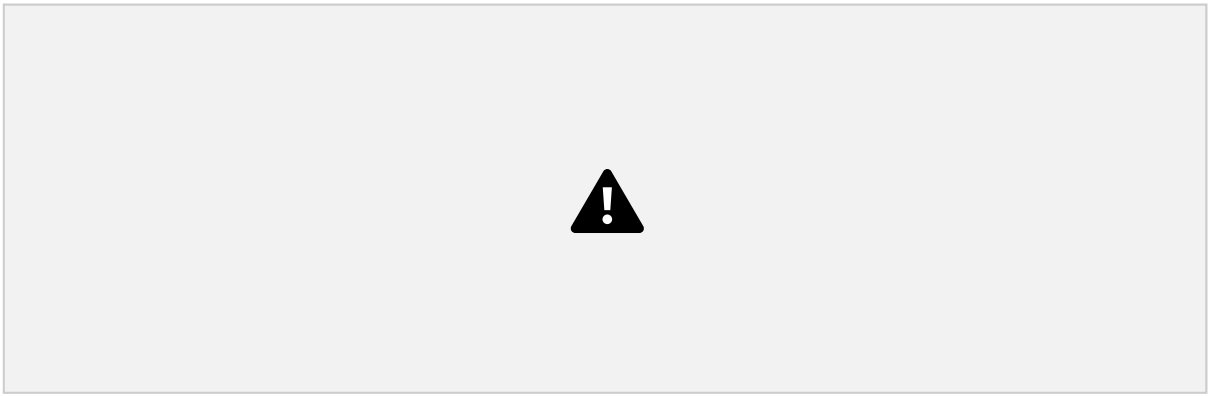




Line chart

• Highest Death Count: January 2024 recorded the highest daily motorist death count, with 5 deaths on a single date. • Low Daily Fatalities Most recorded dates show 0 or 1 motorist death.
• 2025–2026 TrendThe graph shows mostly zero deaths on the recorded dates during 2025 and early 2026





1. Tools and Technologies

Tools / Technologies	Purpose
Python	Programming and data analysis
Pandas	Data cleaning and manipulation
N umPy	Numerical calculations
Matplotlib	Data visualization
Seaborn	Statistical charts and graphs
Google Colab	Development and execution environment
Excel / CSV	Dataset storage and handling

Final Outcome

The project successfully transforms raw motor vehicle crash data into meaningful insights using Python-based data analytics. It helps understand accident trends, vehicle involvement, contributing factors, and injury patterns. These findings can support better road safety awareness, traffic management, and accident prevention strategies.

Conclusion & Recommendations:

- This analysis reveals that traffic accidents in New York City are heavily dictated by peak commuting times, seasonality, and vehicle size disparities.
- Targeted Patrols: Deploying higher law enforcement presence and speed cameras during peak afternoon • Seasonal

Safety Campaigns: Launching public awareness campaigns ahead of the peak accident months in May/June and October/November.

• Strict Safety Standards for SUVs & Sedans: Promoting defensive driving, maintaining safe following distances, and strictly enforcing seatbelt usage to reduce the severity of impacts between larger SUVs and car size is smaller in sedans.

• The analysis highlights the importance of understanding crash frequency across different time periods, identifying common contributing factors

• That support better road safety awareness, informed decision-making, and effective traffic management strategies.